A4.1 Evolution and speciation

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-a4-1.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 138–150 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Evidence for evolution

A4.1.1 Evolution is a change in heritable characteristics

pp. 138–140

Evolution is a change in the inherited characteristics of a population. Changes acquired during a lifetime are not genetic, so they are not evolution. This is what separates Darwin's theory from Lamarck's idea that acquired traits pass to offspring.

A4.1.2 Evidence from sequences

p. 141

Base sequences in DNA or RNA, and amino acid sequences in proteins, can be compared. Mutations accumulate over time, so closer relatives have more similar sequences. This is powerful evidence of common ancestry.

A4.1.3 Evidence from selective breeding

pp. 141–144

Humans have chosen which animals and crops breed. Domesticated breeds now differ greatly from one another and from wild species. This shows how rapidly heritable characteristics can change under selection.

A4.1.4 Evidence from homologous structures

pp. 144–145

The pentadactyl limb

A4.1.4

Region	Forelimb	Hindlimb
One upper bone	Humerus	Femur
Two lower bones	Radius and ulna	Tibia and fibula
Wrist or ankle	Carpals	Tarsals
Digits	Metacarpals and phalanges	Metatarsals and phalanges

Adaptations
Bat: long digits for wings. Whale: short, flat paddle. Horse: one large digit for running.

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Homologous structures have a similar basic structure, but they may have different functions. **Pentadactyl limbs** are the classic example: bones follow one pattern, whether used for flying, swimming or walking.

A4.1.5 Convergent evolution and analogous structures

pp. 145–147

Analogous structures have the same function, but different evolutionary origins. They arise by convergent evolution, as with the wings of birds and insects.

New species

A4.1.6 Speciation splits existing species

pp. 147–148

This is the only way in which new species have appeared. Speciation adds to the total of species alive, and extinction takes away from it.

A4.1.7 Reproductive isolation and differential selection

pp. 149–150

Speciation requires both **reproductive isolation**, which stops gene flow, and **differential selection**, which makes traits diverge. A geographical barrier can isolate populations. The Congo River separates bonobos from chimpanzees, which have diverged because of different conditions on each side.

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